Deep Covariance Alignment for Domain Adaptive Remote Sensing Image Segmentation

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***Abstract*— Unsupervised domain adaptive (UDA) image segmentation has recently gained increasing attention, aiming to improve the generalization capability for transferring knowledge from the source domain to the target domain. However, in high spatial resolution remote sensing image (RSI), the same category from different domains (e.g., urban and rural) can appear to be totally different with extremely inconsistent distributions, which heavily limits the UDA accuracy. To address this problem, in this article, we propose a novel deep covariance alignment (DCA) model for UDA RSI segmentation. The DCA can explicitly align category features to learn shared domain-invariant discriminative feature representations, which enhance the ability of model generalization. Specifically, a category feature pooling (CFP) module is first used to extract category features by combining coarse outputs and deep features. Then, we leverage a novel covariance regularization (CR) to enforce the intracategory features to be closer and the intercategory features to be further separate. Compared with the existing category alignment methods, our CR aims to regularize the correlation between different dimensions of the features, and thus performs more robustly when dealing with divergent category features of imbalanced and inconsistent distributions. Finally, we propose a stagewise procedure to train the DCA to alleviate error accumulation. Experiments on both rural-to-urban and urban-to-rural scenarios of the LoveDA dataset demonstrate the superiority of our proposed DCA over other state-of-the-art UDA segmentation methods. Code is available at https://github.com/Luffy03/DCA.**

***Index Terms*— Deep covariance alignment (DCA), remote sensing image (RSI), semantic segmentation, unsupervised domain adaptive (UDA).**

I. Introduction

REMOTE sensing image (RSI) segmentation aims at assigning the corresponding pixel-wise land-cover type at every image pixel, which plays an increasingly significant role for many applications [1]–[4]. However, the large requirement of labeled training samples and the diverse styles among geographic areas (e.g., urban and rural) heavily limited the development of RSI segmentation. One common solution for the lack and discrepancy of data is the unsupervised domain adaptive (UDA), whose goal is to improve model generalizability from different domains. UDA aims to adapt the models trained from the labeled source domain to the unlabeled target domain, and thus alleviate the lack of annotated training samples, which has been attracting much attention in RSI segmentation task.

Recently, with the success of semantic segmentation methods [5]–[11] based on deep convolutional neural networks (CNNs) [12], [13], UDA segmentation has been rapidly developed. In UDA segmentation, the recent works can be generally divided into two groups, i.e., adversarial training (AT) methods [14]–[17] and self-training (ST) methods [18]–[20]. The AT methods adopt a feature extractor to capture domain-invariant features and perform a discriminator to distinguish them. Tsai *et al.* [14] constructed a multilevel adversarial network (AdaptSeg) to effectively perform output space domain adaptation at different feature levels. Luo *et al.* [15] designed a category-level adversarial network (CLAN) to conduct a more delicate level domain calibration. Wang *et al.* [16] proposed a fine-grained adversarial learning framework (FADA) to align shared features. A transferable normalization (TransNorm) method in [17] was further proposed to improve the transferability in UDA. However, these methods are almost based on generative adversarial network (GAN) [21] for AT, which is difficult to train.

In addition, the ST methods have also been widely used in the UDA segmentation task, which involves the model trained in the source domain to generate pseudo-labels for the target domain and finally fine-tunes the model. Typically, Lian *et al.* [18] proposed pyramid curriculums for cross-domain adaptation (PyCDA) to construct self-motivated pyramid curriculums for UDA. Zou *et al.* [19] further proposed a class-balanced self-training (CBST) strategy to avoid the gradual dominance of large classes in pseudo-label generation and introduced spatial priors to refine the generated pseudo-labels. An instance adaptive self-training (IAST) method is also proposed in [20] to select balanced samples. The ST methods always work in a coarse-to-fine manner, which are usually trained with a stagewise mechanism.

The previous UDA methods have also been applied to the RSI segmentation task [22]–[26]. Most of them aim to preform photometric alignment by GAN [21], which can be classified as AT methods. Despite some promising results of

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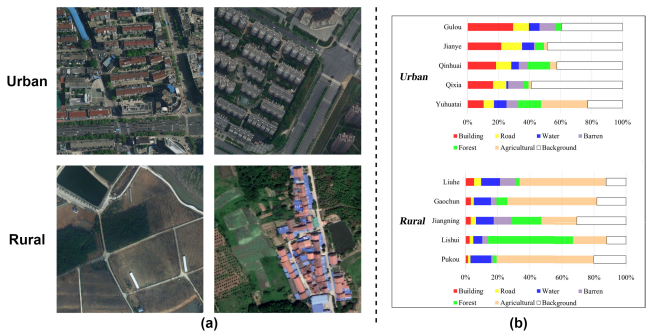


Fig. 1. Difference between urban and rural scenes in the LoveDA dataset [27]. (a) Discrepancy of categories between different domains. (b) Imbalanced and inconsistent category distributions. The details can be seen in [27].

these methods being achieved, actually, the advancements of these algorithms are limited for several reasons. First, as in RSIs, the manifestation of the land-cover is always completely different. Particularly, the same category from diverse areas, i.e., urban and rural, can appear to be totally different in object scales and spectral values. As shown in Fig. 1(a), buildings and roads from urban areas and rural areas are with extremely large discrepancy. Second, the imbalanced and inconsistent category distributions also pose a special challenge for UDA RSI segmentation. As shown in Fig. 1(b), the urban and rural areas have greatly different category distributions, which further increase the difficulty of model generalization in the UDA RSI segmentation task.

To tackle these issues, one direction is to align category-level features between two different domains. The works in [28] and [29] adopt category anchors computed on the source domain to guide the alignment between the two domains, which constrains the category centers with a simple Euclidean distance. The problem of this design is that it does not consider the relationship of intercategory features. Thus, triplet loss is further used in [30] to regularize the distance between intercategory features. Although the above category alignment methods have produced some promising results, for the UDA RSI segmentation task, there are still some urgent issues. First, for RSIs, the larger intracategory variance and the more imbalanced category distributions between two domains will heavily mislead the calculation of distance between different category features. In addition, it is very difficult to learn the decision boundary when adjusting the relative magnitude of intracategory and intercategory simultaneously, which always requires a complex handcrafted threshold setting.

To address the above issues, in this article, we propose a novel deep covariance alignment (DCA) method for UDA RSI segmentation. This model explicitly aligns category features to learn shared domain-invariant discriminative knowledge from the source domain to the target domain. First, a category feature pooling (CFP) module is used to extract category features by combining coarse outputs and deep features. Then, we leverage a novel covariance regularization (CR) to enforce the intracategory features to be closer and the intercategory features to be further separate. Compared with the existing category alignment methods, our CR aims to regularize the correlation between different dimensions of the features, and thus perform more robustly when dealing with the divergent category features of imbalanced and inconsistent distributions. In addition, our CR can be trained without any other complex handcrafted settings. Finally, to alleviate error accumulation, we propose an ST-based stagewise training mechanism for our proposed DCA.

The remainder of this article is organized as follows: Section II reviews the related works. Section III describes the details of our proposed method. Section IV conducts the experiments to verify the effectiveness of our proposed method and comparisons with other methods. Finally, we conclude this article and suggest some future works in Section V.

II. Related Work

A. UDA RSI Segmentation

Recently, UDA RSI segmentation has achieved great progress with several methods proposed [22]–[26]. Most of these methods focus on preforming photometric alignment via GAN [21] to align the source images and the target images in the image space, feature space, and output space. Tasar *et al.* [22] proposed color mapping generative adversarial networks (ColorMapGANs) to generate fake training images for fine-tuning the already trained classifiers. Ji *et al.* [23] further used GAN to align multisource RSIs. However, these AT-based models are hard to train, and they generally align distributions from different domains, but do not actually obtain mappings between features from different domains. Other approaches adopt the idea of ST by generating pseudo-labels for samples in the target domain and providing extra supervision to the classifier [31], [32]. For example, Tong *et al.* [31] proposed an ST-based method for UDA land-cover semantic segmentation, using a transferrable deep model. However, error-prone pseudo-labels will easily mislead the classifier and accumulate errors, which limit the effectiveness of the ST methods.

Despite some promising results being achieved, most of these methods ignore the diverse styles among different geographic areas. While for urban and rural areas, in particular, the manifestation of the land cover is completely different in class distributions, object scales, and pixel spectra. To improve the model generalizability for UDA RSI segmentation, Wang *et al.* [27] originally created a groundbreaking LoveDA dataset, which contains diverse urban and rural RSIs. The dataset aims to promote the capacity of model generalization between the urban domain and the rural domain, which is seriously limited by the large category features' discrepancy and inconsistent category distributions.

B. Category Alignment Methods in UDA

Although the fact that the domain gap can be minimized by the GAN-based AT methods mentioned above, there is no guarantee that features from different categories can be well-separated. Thus, the idea of category-level feature alignment was also exploited in [15] and [33] for UDA segmentation. Other approaches were proposed to match the local joint distributions of features and categories [34], [35]. Category labels

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were further introduced in [36] to enforce global semantic constraints on the distribution of predicted labels. The ideas of minimizing the entropy (uncertainty) of the output [37] have also been exploited to implicitly enforce category-level alignment.

In contrast to the implicit feature alignment in the aforementioned methods, some works [28]–[30] propose to directly align category-wise features in both the domains based on a category anchor-guided method, which have achieved more competitive performance. Among them, [28], [29] adopted category anchors computed on the source domain to guide the alignment between the two domains, which can be regarded as a hard constraint on the category centers. The hard constraint is a distance d between different category features f_1 and f_2, which can be simply defined by a mean square error (MSE) function as

$$d(f_1, f_2) = \|f_1 - f_2\| \tag{1}$$

where $\|\|$ represents a Euclidean distance. f_1 and f_2 represent two features of the same category from two different domains, respectively. This distance d is used to drive the intracategory closer by a loss function L_{mse}

$$L_{\text{mse}}(f_1, f_2) = \frac{1}{N * N} \sum_{i}^{N} d\left(f_1^i, f_2^i\right) \tag{2}$$

where N represents the total number of categories, and f_1^i and f_2^i are the ith category feature from the source domain and the target domain, respectively.

However, this strategy does not explicitly enlarge the margins between the different category features' centers. Thus, Ma *et al.* [30] proposed a category-oriented triplet loss for the source domain that imposes a soft constraint to regularize category centers, actively making intercategory distances in a high-level feature space larger than intracategory distances by a specified margin. The triplet loss L_{triplet} is formulated as follows:

$$L_{\text{triplet}}(f_1, f_2) = \frac{1}{N \times N} \sum_{i}^{N} \sum_{j}^{N} \max\left(d\left(f_1^i, f_2^i\right)\big|_{i=j} - d\left(f_1^i, f_2^i\right)\big|_{i\neq j} + \alpha, 0\right) \tag{3}$$

where α represents a prescribed margin. However, for different categories with large discrepancy, the best margins for training are also different, which requires a well-experienced handcrafted setting. Thus, the work in [30] only applies L_{triplet} to source domain images, which have reliable category labels to supervise.

III. Proposed DCA Method

In this section, we introduce the architecture of our proposed DCA and the detailed training process. First, we introduce the CFP module to extract the category features. Then, we describe the formulation and motivations of our proposed CR and its application to align intradomain and cross-domain features. Finally, we introduce how to train our proposed DCA.

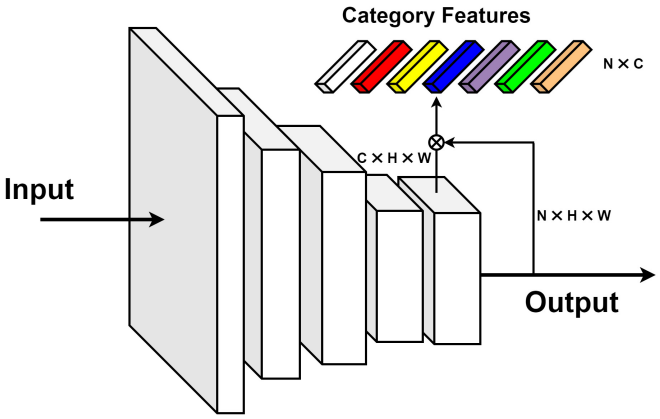


Fig. 2. Framework of the category feature pooling (CFP) module.

A. CFP Module

Specifically, we used Deeplab v2 [38] as the base segmentation model, where ResNet50 [12] is used as the encoder Enc. According to [28], pixels in the same category cluster in the feature space, where the centroids of the features of each category can represent the category feature distributions. Thus, we propose the CFP module to extract the category features from the input image X by combining coarse outputs and deep features. As shown in Fig. 2, we first extract deep features $\text{Enc}(X) \in R^{C\times H\times W}$ and coarse output $Y' \in R^{N\times H\times W}$ from a given input image X, where C and N indicate the number of channels and the number of categories, respectively, and H and W represent the height and width, respectively. Afterward, the category feature $f \in R^{N\times C}$ is calculated as

$$f = \delta(f_1, f_2) \tag{4}$$

where

$$\delta(f_1, f_2) = \frac{1}{H \times W} \sum_{k}^{H\times w} Y'_k \times \text{Enc}(X). \tag{5}$$

Likewise, the category feature f extracted by CFP is also used as the category centroid of the features of each category as a representative of feature distribution. However, it is worth noting that different from the methods in [28]–[30], our f is calculated in only one batch of images instead of all the images in the domain, which produces enormous time savings and an enhancement of efficiency. Also, we do not use label Y to rectify the category feature f as in [28]–[30], since the coarse output Y' has already been supervised by the reliable label Y.

B. Covariance Regularization (CR)

1) Motivations of CR: To align the category features, we hope that the intracategory features can be closer and the intercategory features can be further apart. Given two category features f_1 and f_2 extracted by CFP module, we propose a CR method to align the features, which is defined as

$$\text{Corr}(f_1, f_2) = \frac{E\left[\left(f_1 - \mu_{f_1}\right)\left(f_2 - \mu_{f_2}\right)\right]}{\sigma_{f_1}\sigma_{f_2}} \tag{6}$$

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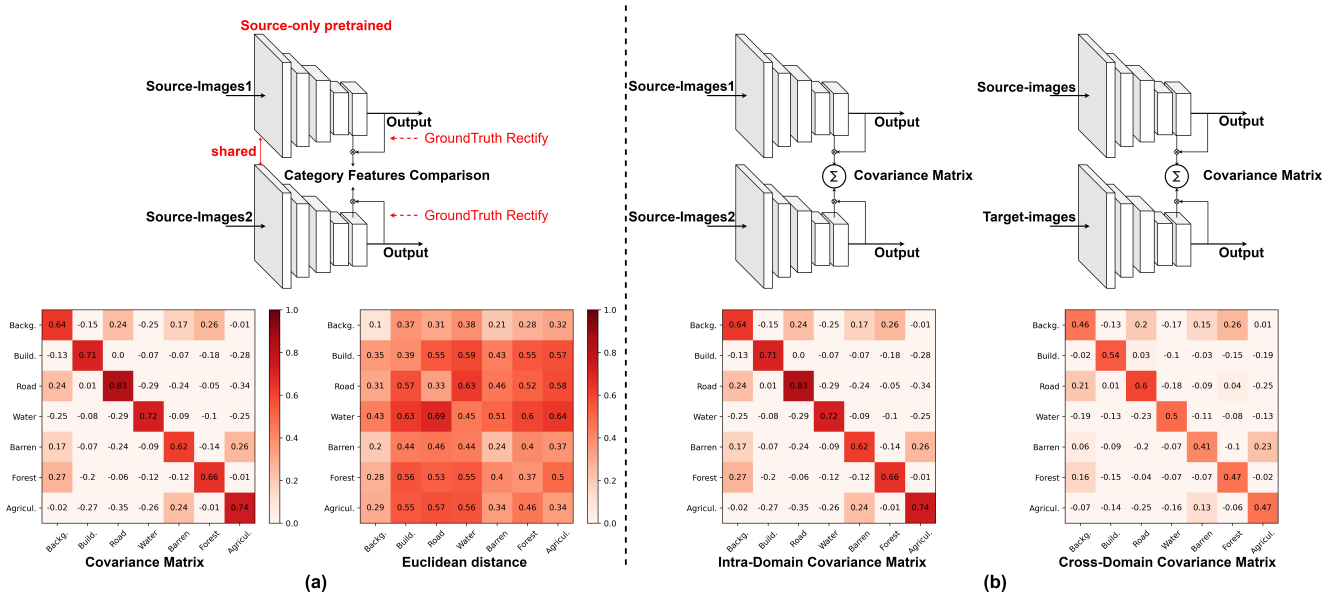


Fig. 3. Explainable experiment for the proposed CR. (a) Comparison between our proposed covariance matrix and the Euclidean distance used in [28], [29], and [30]. (b) Discrepancy of category features from different domain, which is represented by our proposed covariance matrix. It is worth noting that the shared source-only pretrained network is used in (a) and (b). Ground-truth rectify is not used in the actual training as described in Section III-A.

where σ_{f_1} and σ_{f_2} are the average values of f_1 and f_2, respectively, and μ_{f_1} and μ_{f_2} represent the variance of f_1 and f_2, respectively. E represents the expectation function. $\mathrm{Corr}(f_1, f_2)$ builds a $N \times N$ covariance matrix to represent the relationship between different category features. $\mathrm{Corr}(f_1, f_2)$ is also known as the linear correlation coefficient in the field of mathematics, which is a Mahalanobis distance, instead of the Euclidean distance used in the aforementioned methods [28]–[30].

To further illustrate our motivations of CR, we conduct an explainable experiment for the proposed CR as shown in Fig. 3. In Fig. 3(a), we divide two groups of images from the source domain and input them through a source-only pretrained network. Then the category features comparisons are conducted for f_1 and f_2 extracted by the CFP modules. It can be seen in Fig. 3(a) that compared with the Euclidean distance using in [28]–[30], our proposed CR can represent the relationship between different category features much more explicitly. In addition, as shown in Fig. 3(b), we input the source images and the target images into the network to calculate the covariance matrix. We can see in Fig. 3(b) that the discrepancy between the source domain and the target domain can be clearly represented by the covariance matrix. With a covariance matrix, we hope that the diagonal elements are close to 1 and the off-diagonal elements are less than 0, and thus CR is formulated as

$$L_{\mathrm{CR}}(f_1, f_2) = \frac{1}{N \times N} \sum_{i}^{N} \sum_{j}^{N} \log\big(A_{ij}(\mathrm{Corr}(f_1, f_2))\big) \quad (7)$$

where

$$A_{ij}(\mathrm{Corr}) = \begin{cases} \mathrm{Corr}, & i = j \\ \max(1 - \mathrm{Corr}, \epsilon), & i \neq j \end{cases} \quad (8)$$

where ϵ is a small value to avoid logarithm with zero.

We summarize the novelty of CR from three aspects. First, as shown in Fig. 3, our proposed CR is more robust when calculating the distance between category features. Since the large intracategory variance in RSIs will mislead the calculation of category features, the Euclidean distance used in [28], [29], and [30] cannot represent the category distance distinctly. But according to (6), our CR is normalized with the variance σ_f, thus alleviating the impact of intracategory variance. In addition, with the extremely imbalanced and inconsistent category distributions from two domains, the calculation of Euclidean distance almost depends on the category with larger proportion, which cannot regularize the category features effectively. However, as a kind of Mahalanobis distance, our CR is also scale-invariant and not affected by dimensions, which means the category with small proportion can also impact the calculation of distance heavily. Thus, for RSIs with larger intracategory variance and inconsistent category distributions, our proposed CR can better align the category features between two domains than the existing category alignment methods.

Second, unlike the Euclidean distance that only depends on the value difference, our CR concerns more about the relationship between different dimensions of the vectors. With high-dimension category features extracted, the existing methods [28]–[30] use the Euclidean distance to impose a hard constraint to regularize category features, but ignore the relationship between different dimensions. Instead, our CR encourages the model to extract linear-correlated or linear-independent features. For example, for the same categories, the CR enforces the same dimensions of features to be stimulated in the model, while invariant dimensions of features are stimulated for different categories. Thus, our CR can align the category features from dimension aspects, instead of the unstably calculated value difference.

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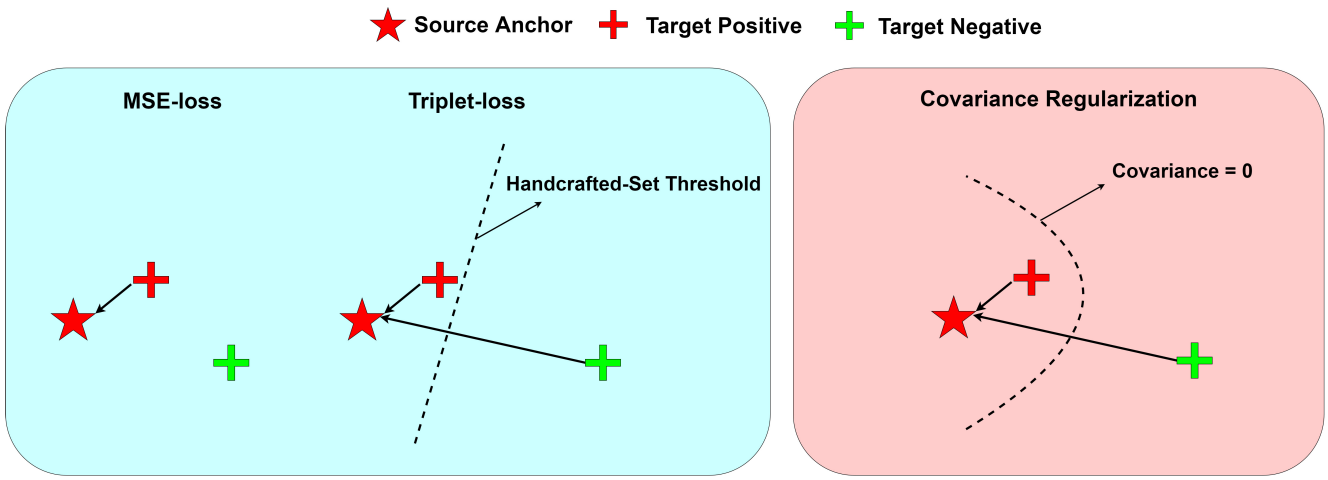


Fig. 4. Comparisons between our proposed CR and the existing category alignment methods.

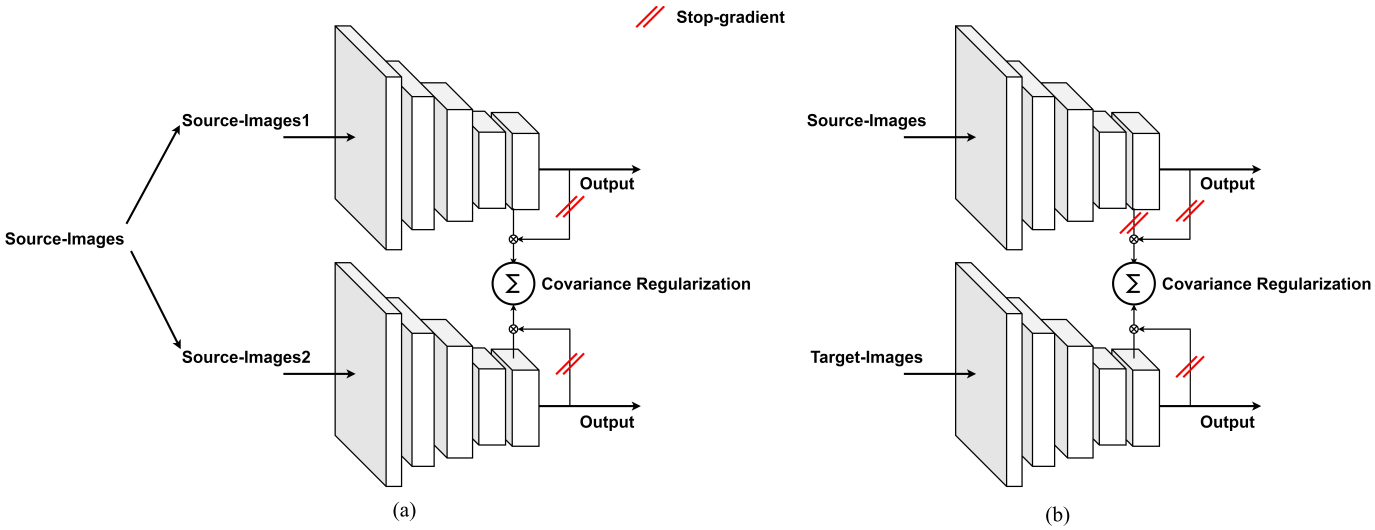


Fig. 5. Architectures of the proposed (a) ICR and (b) CCR.

Finally, our proposed CR can adapt the UDA model more effectively compared with the existing category alignment methods. As shown in Fig. 4, our CR has outstanding advantages compared with the methods in [28]–[30]. First, the MSE used in [28], [29] does not align different category features. Although the triplet proposed in [30] can regularize different category features, it is very difficult to learn the decision boundary when adjusting the relative magnitude of intracategory and intercategory simultaneously, which always requires a complex handcrafted threshold setting according to (3). However, our CR can not only align different category features but also save the step to set the threshold, since corvariance $=$ 0 is a native threshold. With corvariance $>$ 0, the category features are linear-correlated, while for corvariance $\leq$ 0 they are linear-independent or negative-correlated. Thus, corvariance $=$ 0 is the best decision boundary for aligning intracategory and intercategory features, which does not require any other handcrafted settings.

2) Intradomain CR and Cross-Domain CR: As shown in Fig. 5(a), for ICR, given a batch of images, we first divide them into two groups randomly, and then we perform CR to regularize the two groups of category features. Inspired by [39], in ICR, we stop the gradient for outputs to optimize the learning of deep representations only, since the outputs are with reliable supervision from labels. It is worth noting that ICR is only used in the source domain, since the coarse outputs in the source domain are supervised by reliable labels, which are much more accurate. The regularization loss L_{ICR} is denoted as

$$L_{\text{ICR}} = L_{\text{CR}}\left(f_s^1, f_s^2\right) \tag{9}$$

where f_s^1 and f_s^2 are the category features from different groups in the source domain, respectively.

CCR is further used to align the category features between different domains. While in CCR, category features from the source domain and outputs from both the domains are fixed, the category features captured from target images are optimized only. Since the source domain is with more correct supervision, we assign them as credible reference, and we can align the target features to the source features. Similarly, L_{CCR}

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TABLE I
PERFORMANCE COMPARISONS WITH BASELINE METHODS AND CATEGORY ALIGNMENT METHODS

Domain	Method	IoU (%)							mIoU (%)
		Background	Building	Road	Water	Barren	Forest	Agriculture	
Rural-To-Urban	Source-only	42.45	26.76	25.08	70.63	12.98	17.97	26.83	31.86
	Baseline	39.76	43.78	33.91	69.65	8.69	41.71	22.67	37.17
	+MSE [27]	44.79	42.12	34.97	79.97	16.06	31.44	36.59	40.85
	+Triplet [30]	40.69	43.78	37.09	80.76	16.27	20.78	32.85	38.89
	DCA	**45.82**	**49.60**	**51.65**	**80.88**	**16.70**	**42.93**	**36.92**	**46.36**
Urban-To-Rural	Source-only	25.27	44.01	22.64	54.30	6.97	31.00	39.93	32.02
	Baseline	26.72	45.87	25.71	58.46	7.36	38.73	45.82	35.52
	+MSE [27]	26.49	46.28	34.19	60.45	1.15	38.03	54.65	37.60
	+Triplet [30]	27.65	45.12	25.03	58.68	11.22	37.93	50.62	36.61
	DCA	**36.38**	**55.89**	**40.46**	**62.03**	**22.01**	**38.92**	**60.52**	**45.17**

[1] Note: **+MSE** and **+Triplet** means employ MSE based alignment [28] and Triplet based alignment [30] with the Baseline, respectively. And Baseline is the common ST strategy described in Section IV.C(1).

TABLE II
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART UDA SEGMENTATION METHODS IN RURAL-TO-URBAN EXPERIMENT

Method	Type	IoU (%)							mIoU (%)
		Background	Building	Road	Water	Barren	Forest	Agriculture	
Source-only	-	42.45	26.76	25.08	70.63	12.98	17.97	26.83	31.86
MCD [42]	-	43.60	15.37	11.98	79.07	14.13	33.08	23.47	31.53
AdaptSeg [14]	AT	42.35	23.73	15.61	81.95	13.62	28.70	22.05	32.68
FADA [15]	AT	43.89	12.62	12.76	80.37	12.70	32.76	24.79	31.41
CLAN [16]	AT	43.41	25.42	13.75	79.25	13.71	30.44	25.80	33.11
TransNorm [17]	AT	38.37	5.04	3.75	80.83	14.19	33.99	17.91	27.73
PyCDA [18]	ST	38.04	35.86	45.51	74.87	7.71	40.39	11.39	36.25
CBST [19]	ST	48.37	46.10	35.79	80.05	19.18	29.69	30.05	41.32
IAST [20]	ST	**48.57**	31.51	28.73	**86.01**	**20.29**	31.77	36.50	40.48
DCA	ST	45.82	**49.60**	**51.65**	80.88	16.70	**42.93**	**36.92**	**46.36**

is defined as

$$L_{\mathrm{CCR}} = L_{\mathrm{CR}}(f_s, f_t) \tag{10}$$

where f_s and f_t are the category features from the source domain and the target domain, respectively.

With the ICR and the CCR, we encourage the intracategory features closer and the intercategory features further separate in both the domains, which adapts the proposed DCA model more progressively.

C. Training of DCA

Similar to the aforementioned ST methods, we train the DCA model with a stagewise procedure. The training procedure is proposed to avoid accumulation of error-prone pseudo-labels generated in ST, which will produce incorrect supervision signals, leading to more erroneous pseudo-labels iteratively and trap the network to a local minimum with poor performance eventually. The procedure is shown in Algorithm 1. First, we pretrain the segmentation model [38] on the source domain. A standard cross-entropy loss function $\mathrm{CE}(Y', Y)$ is used to supervise as

$$L_{\mathrm{CE}}^{s} = \mathrm{CE}\big(Y_s', Y_s\big) \tag{11}$$

where

$$\mathrm{CE}\big(Y', Y\big) = -\frac{1}{H \times W} \sum_{k}^{H \times W} \sum_{i}^{N} \big[Y_{ki} \log\big(Y_{ki}'\big) + (1 - Y_{ki}) \log\big(1 - Y_{ki}'\big)\big]. \tag{12}$$

Algorithm 1 Stagewise Training of the DCA Model

Data: training dataset: (X_s, Y_s, X_t), maximum stages: K, maximum iterations I
Result: Model M_K and (Y_s', Y_t')

```
while i ≤ I do
    if k = 0 then
        Pretraining: M_0 ← (X_s, Y_s);
        ICR: L_ICR ← (X_s, Y_s') according to (9);
    end
    for k ← 1 to K do
        Generate Y_t^p ← M_{k−1};
        ICR: L_ICR ← (X_s, Y_s') according to (9);
        CCR: L_CCR ← (X_s, Y_s', X_t, Y_t') according to (10);
        Training M_k on (X_s, Y_s, X_t, Y_t^p, L_ICR, L_CCR)
        according to (14)
    end
    M_{k−1} ← M_k
end
```

Meanwhile, the ICR is used to align the category features in the source domain. At the beginning of each stage, we generate or update the pseudo-labels Y_t^p to supervise the target predictions Y_t' as

$$L_{\mathrm{CE}}^{t} = \mathrm{CE}\big(Y_t', Y_t^p\big). \tag{13}$$

Next, we trained the DCA with L_{CE}^{s}, L_{CE}^{t} and the proposed L_{ICR} and L_{CCR} for several stages

$$L = L_{\mathrm{CE}}^{s} + L_{\mathrm{CE}}^{t} + L_{\mathrm{ICR}} + L_{\mathrm{CCR}}. \tag{14}$$

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TABLE III

PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART UDA SEGMENTATION METHODS IN URBAN-TO-RURAL EXPERIMENT

Method	Type	IoU (%)							mIoU (%)
		Background	Building	Road	Water	Barren	Forest	Agriculture	
Source-only	-	25.27	44.01	22.64	54.30	6.97	31.00	39.93	32.02
MCD [42]	-	25.61	44.27	31.28	44.78	13.74	33.83	25.98	31.36
AdaptSeg [14]	AT	26.89	40.53	30.65	50.09	16.97	32.51	28.25	32.27
FADA [15]	AT	24.39	32.97	25.61	47.59	15.34	34.35	20.29	28.65
CLAN [16]	AT	22.93	44.78	25.99	46.81	10.54	37.21	24.45	30.39
TransNorm [17]	AT	19.39	36.30	22.04	22.04	**36.68**	14.00	40.62	24.62
PyCDA [18]	ST	12.36	38.11	20.45	57.16	18.32	36.71	41.90	32.14
CBST [19]	ST	25.06	44.02	23.79	50.48	8.33	**39.16**	49.65	34.36
IAST [20]	ST	29.97	49.48	28.29	**64.49**	2.13	33.36	**61.37**	38.44
DCA	ST	**36.38**	**55.89**	**40.46**	62.03	22.01	38.92	60.52	**45.17**

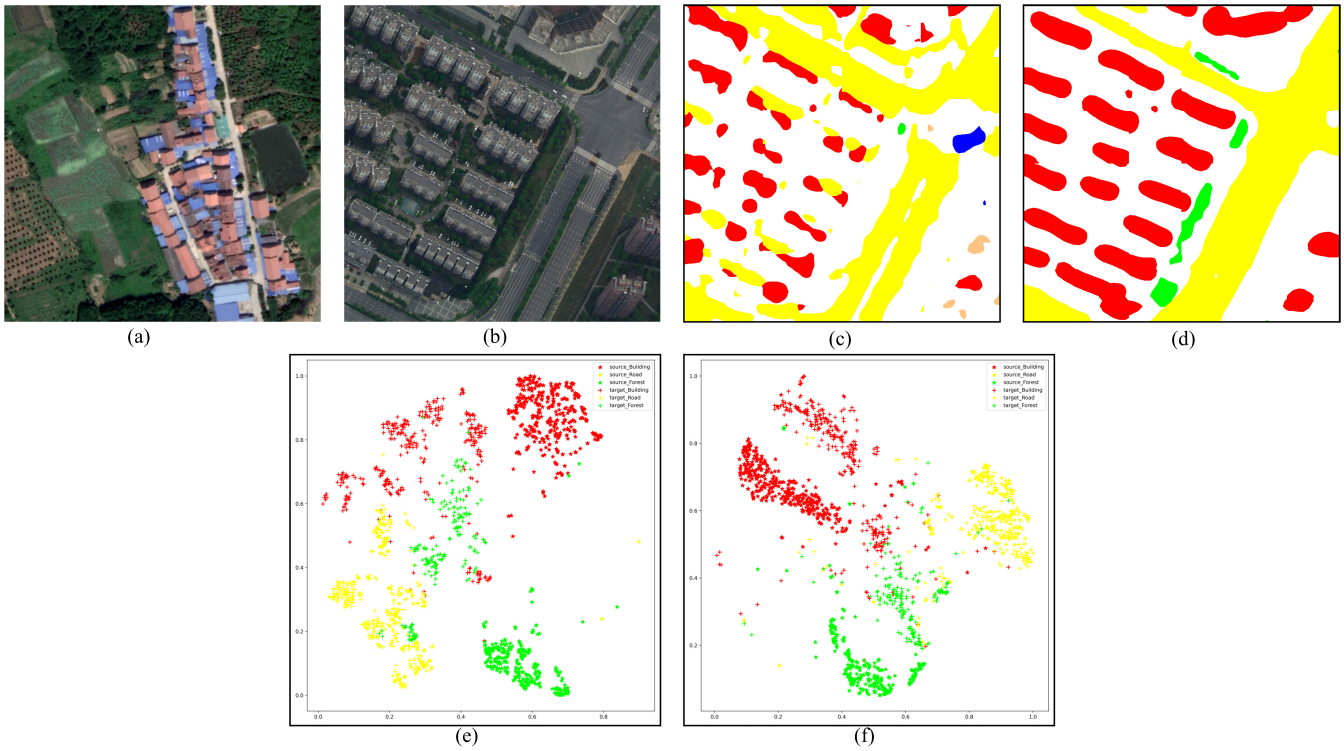


Fig. 6. Contrastive analysis of the proposed DCA and the baseline method on the rural-to-urban experiment. (a) Source image from the rural domain. (b) Target image from the urban domain. (c) (d) Segmentation results of the baseline method and our DCA method, respectively. We then map the high-dimension features of (c) and (d) to a 2-D space with t-stochastic neighbor embedding (SNE) [40], shown in (e) and (f).

IV. RESULTS OF EXPERIMENTS

To verify the effectiveness of our proposed method, we conducted extensive experiments on the advanced UDA RSI segmentation dataset LoveDA [27]. In this section, we first introduce the dataset and implementation details. Then, we perform detailed extensive ablation experiments on the dataset. Finally, we report our results on both the rural-to-urban and urban-to-rural scenarios, including the visual results from different methods.

A. Datasets

The LoveDA dataset contains 5987 high spatial resolution images (1024 × 1024 pixels) from three different cities. Compared with the existing datasets, the LoveDA dataset encompasses two domains (urban and rural), which focuses on improving the generalization capability of model from different urban and rural scenes. For the rural-to-urban task, there are 1366 source images (rural) and 677 target images (urban) for training and 820 target images (urban) for testing. As for the urban-to-rural task, 1156 source images (urban) and 992 target images (rural) are used for training, while 976 target images (rural) for testing. The detailed data splits are described in [27].

B. Experimental Settings

1) Experimental Environment: All the experiments were processed on a server computer with an NVIDIA A100 GPU. The implementation of the framework was based on the open-source toolbox Pytorch. Each experiment is conducted for five times, and the average results are reported.

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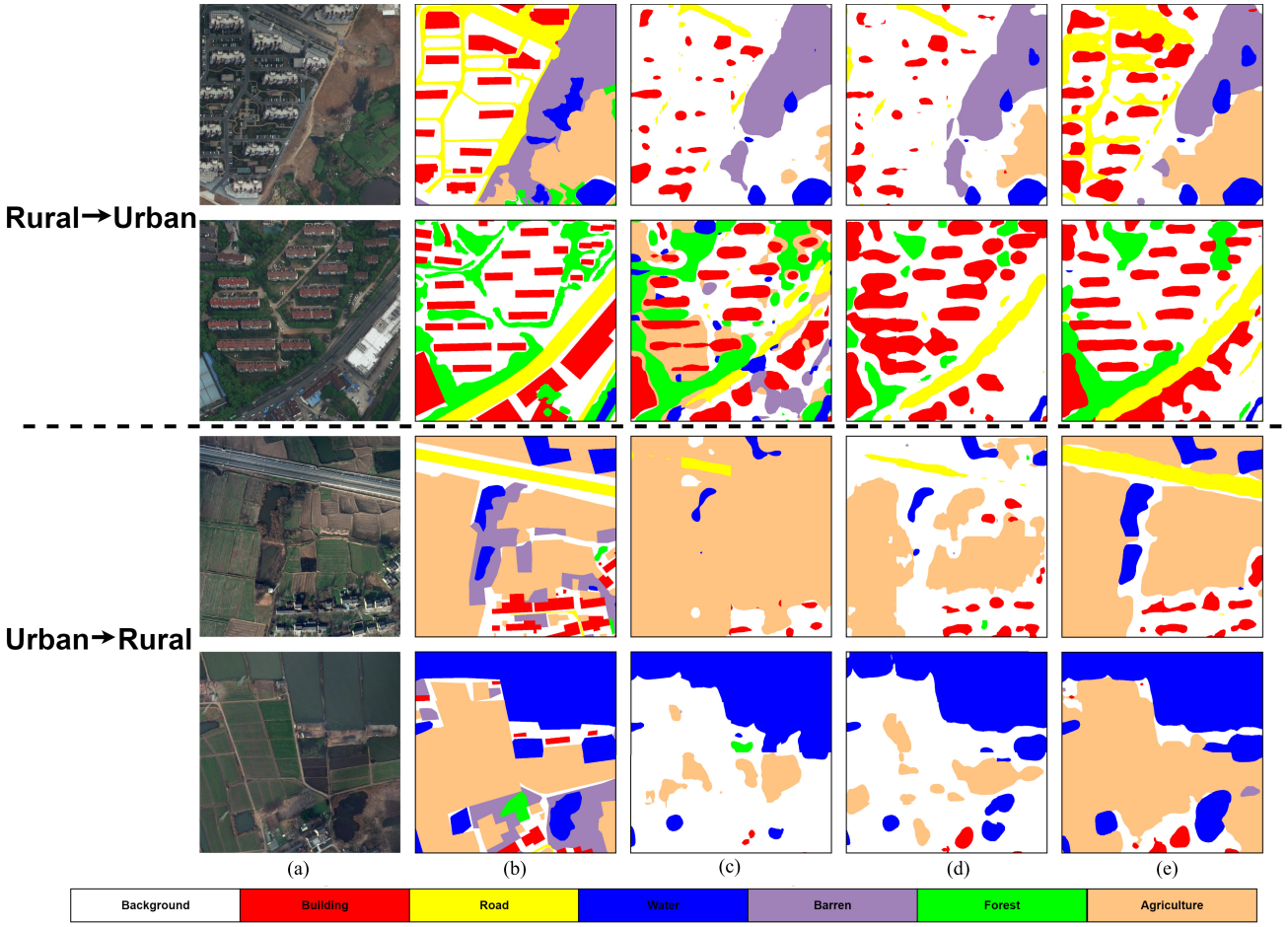


Fig. 7. Qualitative comparisons between the proposed methods and other methods on the Val dataset. (a) Image. (b) Ground truth. (c) Source only. (d) CBST. (e) DCA.

2) Implementation Details: We use intersection over union (IoU) to report the semantic segmentation accuracy. With respect to IoU for each class, mIoU represents the mean of IoUs over all the categories.

All the methods adopt the same feature extractor and discriminator, Specifically, DeepLabV2 [38] with ResNet50 [12] was used as the extractor, and the discriminator was constructed by fully convolutional layers [14]. The backbones used in all the networks were pretrained on ImageNet [41]. According to the experiments in [27], during training, we used the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and a weight decay of 0.0005. The learning rate was initially set to 0.01, and a poly schedule with power 0.9 was applied. For data augmentation, 512 × 512 patches were randomly cropped from raw images (1024 × 1024 pixels) for training, with random mirroring and rotation. While in the testing part, we perform slide predictions for raw images, to alleviate the resolution's impact on segmentation results. In the stagewise training Algorithm 1, the maximum stage K was set to 5 and the maximum iteration I was set to 10 000 for both the experiments.

C. Ablation Study for DCA

1) Comparisons With Category Alignment Methods: To validate the efficiency of our DCA, we first compare our DCA method with the source-only method and the baseline method. Here, we define the baseline method as using the common ST strategy, which simply pretrained the model in the source domain and generated the pseudo-labels to supervise the target images according to (13) directly without any other prior. Next, we further conducted a fair comparison between the proposed CR and other category alignment methods, i.e., MSE-based alignment in [28] and triplet-based alignment in [30]. As for the triplet alignment method, we tried several times to find the best α according to (3) and finally set it to 0.5. The detailed results are shown in Table I.

As shown in Table I, the ST baseline can promote the generalization of the source-only pretrained model by a large margin. In addition, the MSE alignment method brings improvements in mIoU, which illustrates the effectiveness of category alignment methods. However, the triplet alignment method is not effective in enhancing the accuracy. The reason may be that the training of triplet is complex and not available for UDA RSI segmentation. In addition, our proposed DCA can achieve a remarkable improvement compared with other methods. In the rural-to-urban experiment, our DCA yields a 9.19% improvement in mIoU and 9.65% improvement in mIoU for the urban-to-rural experiment, which outperforms the other methods with a large margin. This demonstrates that our approach can greatly improve the generalization capability of model by aligning category features.

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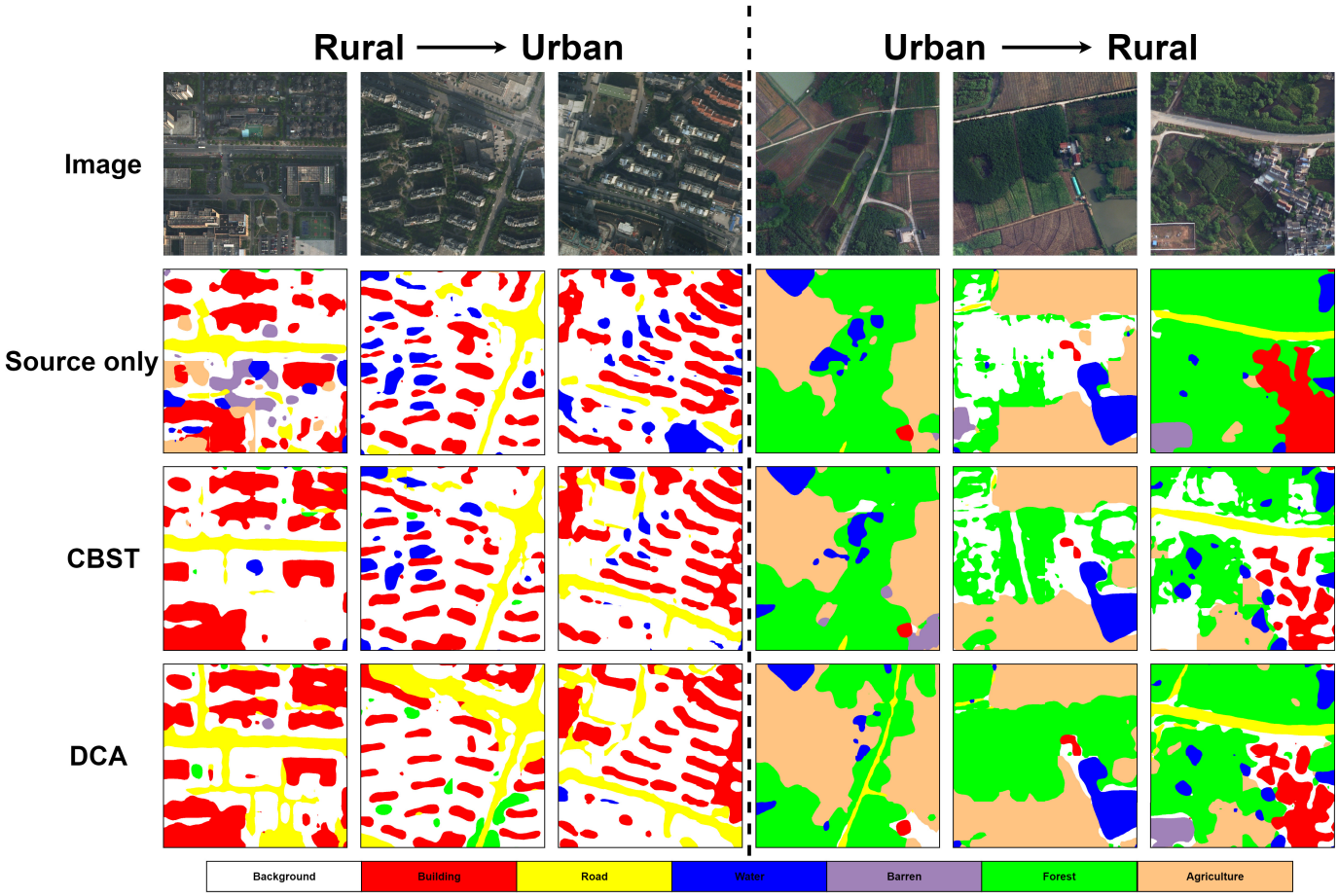


Fig. 8. Qualitative comparisons between the proposed methods and other methods on the test dataset.

2) Visualization Results of Category Features' Distributions: To further illustrate the novelty of our proposed DCA, we visualize deep category features' distributions in latent space. To this end, given a source image and a target image with large domain discrepancy, we map their high-dimensional deep features to a 2-D space with t-SNE [40], as shown in Fig. 6. We focus on the feature distributions between different categories from two domains. As shown in Fig. 6(e), for the baseline method, the learned features of buildings, roads, and forests are separate due to the domain shift. It can be seen that without CR alignment, the baseline method cannot explicitly learn the shared representations of the same categories from different domains. Instead, we can observe in Fig. 6(f) that our proposed DCA can drive the intracategory features closer and intercategory features further apart, which build more accurate UDA segmentation results shown in Fig. 6(d).

3) Comparisons With UDA Segmentation Methods: The LoveDA dataset contains two scenarios: rural-to-urban and urban-to-rural. We carry out experiments for both the scenarios on the LoveDA dataset [27] to evaluate the effectiveness of our DCA compared with other state-of-the-art UDA segmentation methods, as shown in Tables II and III.

It can be seen from Table II that in the rural-to-urban experiments, the DCA achieves the highest mIoU of 46.36%, which outperforms the state-of-the-art method CBST by 5.04%. Specifically, the DCA brings a great improvement of IoU of buildings, roads, and forests. However, for background and barren, the IoUs do not improve obviously. The reason may be that the features of these two categories are too complex to align. For the urban-to-rural experiments shown in Table III, DCA also achieves an mIoU of 45.17% with a large gap compared with other methods. The improvements are mainly from buildings and roads, which means our DCA can effectively extract the shared features of these categories from the rural domain to urban domain. The results demonstrate the efficiency of our proposed DCA method.

4) Visualization Results: The qualitative results are shown in Figs. 7 and 8. Since we do not have the corresponding ground truth on the test dataset, we also display the results on the Val dataset in Fig. 7 to compare with the ground truths. Fig. 8 shows the visualization results on the test dataset.

It can be seen that our DCA can produce better UDA segmentation predictions than the source-only method and CBST [19]. Specifically, in the rural-to-urban experiments, our DCA predicts more complete constructions for buildings and roads. As in the urban-to-rural experiments, the segmentation results of agriculture are more accurate. The visualization results can also demonstrate the effectiveness of our proposed framework.

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V. Conclusion

In this article, we presented a novel DCA framework for UDA RSI segmentation. Our proposed DCA used a novel CR to enforce the intracategory features to be closer and the intercategory features to be further separate, and thus explicitly extracted shared domain-invariant discriminative feature representations to enhance the ability of model generalization. Extensive experiments on the LoveDA dataset demonstrated the effectiveness and efficiency of the proposed DCA compared with other state-of-the-art methods.

In the future, we will explore the development of quantitative experiments to analyze visualization and interpretation of results performed by our DCA and compared with other approaches. Moreover, we will investigate the proficiency of the domain adaptive segmentation task in RSIs and consider other alignment approaches to further improve segmentation performance.

References

[1] D. Li, G. Zhang, Z. Wu, and L. Yi, "An edge embedded marker-based watershed algorithm for high spatial resolution remote sensing image segmentation," *IEEE Trans. Image Process.*, vol. 19, no. 10, pp. 2781–2787, Oct. 2010.

[2] H. Sheng, X. Chen, J. Su, R. Rajagopal, and A. Ng, "Effective data fusion with generalized vegetation index: Evidence from land cover segmentation in agriculture," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 267–276.

[3] L. Ding, J. Zhang, and L. Bruzzone, "Semantic segmentation of large-size VHR remote sensing images using a two-stage multiscale training architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5367–5376, Aug. 2020.

[4] Q. Wang, J. Gao, and X. Li, "Weakly supervised adversarial domain adaptation for semantic segmentation in urban scenes," *IEEE Trans. Image Process.*, vol. 28, no. 9, pp. 4376–4386, Sep. 2019.

[5] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[6] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, vol. 9351. Cham, Switzerland: Springer, Nov. 2015, pp. 234–241.

[7] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Jun. 2020.

[8] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[9] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. CVPR*, Jul. 2017, pp. 936–944.

[10] J. Xie, L. Fang, B. Zhang, J. Chanussot, and S. Li, "Super resolution guided deep network for land cover classification from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[11] N. He, L. Fang, and A. Plaza, "Hybrid first and second order attention UNet for building segmentation in remote sensing images," *Sci. China Inf. Sci.*, vol. 63, no. 4, pp. 1–12, Apr. 2020.

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jan. 2016, pp. 770–778.

[13] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[14] Y.-H. Tsai, W.-C. Hung, S. Schulter, K. Sohn, M.-H. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. CVPR*, Jun. 2018, pp. 7472–7481.

[15] Y. Luo, L. Zheng, T. Guan, J. Yu, and Y. Yang, "Taking a closer look at domain shift: Category-level adversaries for semantics consistent domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2502–2511.

[16] H. Wang, T. Shen, W. Zhang, L. Duan, and T. Mei, "Classes matter: A fine-grained adversarial approach to cross-domain semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 642–659.

[17] X. Wang, Y. Jin, M. Long, J. Wang, and M. I. Jordan, "Transferable normalization: Towards improving transferability of deep neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Jul. 2019, pp. 1–11.

[18] Q. Lian, L. Duan, F. Lv, and B. Gong, "Constructing self-motivated pyramid curriculums for cross-domain semantic segmentation: A non-adversarial approach," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6757–6766.

[19] Y. Zou, Z. Yu, B. V. Kumar, and J. Wang, "Unsupervised domain adaptation for semantic segmentation via class-balanced self-training," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 289–305.

[20] K. Mei, C. Zhu, J. Zou, and S. Zhang, "Instance adaptive self-training for unsupervised domain adaptation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 415–453.

[21] I. Goodfellow *et al.*, "Generative adversarial nets," 2014, *arXiv:1406.2661*.

[22] O. Tasar, S. L. Happy, Y. Tarabalka, and P. Alliez, "ColorMapGAN: Unsupervised domain adaptation for semantic segmentation using color mapping generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7178–7193, Oct. 2020.

[23] S. Ji, D. Wang, and M. Luo, "Generative adversarial network-based full-space domain adaptation for land cover classification from multiple-source remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 3816–3828, May 2021.

[24] L. Yan, B. Fan, H. Liu, C. Huo, S. Xiang, and C. Pan, "Triplet adversarial domain adaptation for pixel-level classification of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3558–3573, May 2020.

[25] O. Tasar, A. Giros, Y. Tarabalka, P. Alliez, and S. Clerc, "DAugNet: Unsupervised, multisource, multitarget, and life-long domain adaptation for semantic segmentation of satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1067–1081, Feb. 2021.

[26] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[27] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, vol. 1. J. Vanschoren and S. Yeung, 2021, pp. 1–12. [Online]. Available: https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/file/4e732ced3463d06de0ca9a15b6153677-Paper-round2.pdf

[28] Q. Zhang, J. Zhang, W. Liu, and D. Tao, "Category anchor-guided unsupervised domain adaptation for semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Oct. 2019, pp. 433–443.

[29] Z. Wang *et al.*, "Differential treatment for stuff and things: A simple unsupervised domain adaptation method for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12632–12641.

[30] H. Ma, X. Lin, Z. Wu, and Y. Yu, "Coarse-to-fine domain adaptive semantic segmentation with photometric alignment and category-center regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4050–4059.

[31] X.-Y. Tong *et al.*, "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[32] A. Zheng, M. Wang, C. Li, J. Tang, and B. Luo, "Entropy guided adversarial domain adaptation for aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[33] K. Saito, Y. Ushiku, T. Harada, and K. Watanabe, "Maximum classifier discrepancy for unsupervised domain adaptation," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 3723–3732.

[34] Y.-H. Chen, W.-Y. Chen, Y.-T. Chen, B.-C. Tsai, Y.-C.-F. Wang, and M. Sun, "No more discrimination: Cross city adaptation of road scene segmenters," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2011–2020.

[35] G. Kang, L. Jiang, Y. Yang, and A. G. Hauptmann, "Contrastive adaptation network for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4888–4897.

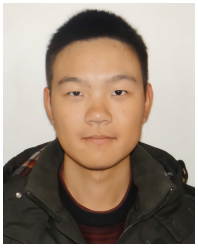
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[36] J. Hoffman, D. Wang, F. Yu, and T. Darrell, "FCNs in the wild: Pixel-level adversarial and constraint-based adaptation," 2016, *arXiv:1612.02649*.
[37] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Perez, "ADVENT: Adversarial entropy minimization for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2512–2521.
[38] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[39] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15745–15753.
[40] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.
[41] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2009, pp. 248–255.
[42] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell, "Deep domain confusion: Maximizing for domain invariance," 2014, *arXiv:1412.3474*.



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